Gowrynanda PS

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EDUCATION

Indian Institute of Technology Madras, Chennai, India — *Expected Graduation: May 2026*
B.Tech(Hons.) in Mechanical Engineering + M.Tech in Robotics — ***CGPA: 9.01/10***

Christ Nagar Senior Secondary School, Thiruvallam, India — *2021*
Class XII; — *Percentage: 99.2%*

Indian School Al Seeb, Muscat, Oman — *2019*
Class X; — *Percentage: 99%*

PATENTS AND PUBLICATIONS

Mass Afforestation Drone- (*Patent Number - 549953*)

- Designed and developed a drone with seed storage, sorting, and planting capabilities, equipped with a **novel suspension system** to maintain stability during soil interaction and a soil scooper engineered to minimize destabilizing forces

RESEARCH EXPERIENCE

Thesis – Trajectory Planning and Control of a Low-Cost Quadruped Leg Design
May 2025 – Current

Guide: Dr. Bijo Sebastian, Autonomous Systems Lab, IIT Madras

- Built a **quadruped leg prototype** using a novel **rotary series elastic actuator** for greater impact tolerance and torque output.
- Developed a complete testbed with **dynamics and control framework** in CoppeliaSim with torque-level actuation
- Currently integrating **terrain perception** using Koopman framework for sensor-less terrain detection and adaptive locomotion.

Low-Inertia Cable-Driven Underwater Manipulator
Oct 2024 – Current

Guides: Dr. Abilash Somayajula and Dr. Bijo Sebastian – Industry-collaborative project with Oceaneering, Houston, USA

- Designed a **cable-driven underwater manipulator** with optimized pulley routing to minimize distal loading
- Developed and simulated **Optimal Control strategies** to enhance accuracy and stability under underwater disturbances
- Demonstrated robust performance through **gravity-compensated PID control** validated using dynamic simulations

Smart Compression Garment for Hypotension Prevention in Paraplegic Individuals
May 2023 – Mar 2024

Guide: Dr. Manish Anand, TTK Center for Rehabilitation Research and Device Development (R2D2), IIT Madras

- Integrated **force resistive sensors** for real-time pressure monitoring in the compression garment
- Developed a **feedback control system** to dynamically regulate pressure, preventing pressure sores and improving blood circulation

Autonomous Afforestation Drone with a Robotic Manipulator
May 2022 – Apr 2023

Guide: Dr. Manish Anand, IIT Madras – Center for Innovation (CFI)

- Designed and fabricated a lightweight **robotic arm** in Fusion 360 for soil analysis and precision seed sowing in favourable locations
- Integrated the manipulator with a **hexacopter platform**, enabling autonomous flight and target-site positioning using ardupilot
- Built a controlled testbed to validate sowing accuracy under variable terrain and soil moisture conditions

RELEVANT COURSE PROJECTS

Lyapunov Based Control Barrier Functions for Safe 3D Navigation
Jan 2026– Present

Course: Non-Linear Systems Analysis by Prof.Puduru Viswanadha Reddy, IIT Madras

- Developed a spatially shifted CBF framework using a single CLF and translational symmetry to synthesize multiple CBF online
- Implemented an autonomous switching mechanism with a real time CBF-QP optimization reducing online iteration time to 30 ms
- Validated safety and goal convergence on a 10 state quadrotor model in a complex 3D simulation

Data driven terrain detection for quadruped locomotion
July 2025 – Nov 2025

Course: Data Driven Control by Prof.Puduru Viswanadha Reddy, IIT Madras

- Used a data driven approximation of the leg dynamics as a **switched dynamic system** differentiating swing and stance phases
- Simulated different terrain behaviors and found the approximate **Koopman generators** from the data collected
- Performed an **eigen analysis** of the stance generators to accurately predict the **terrain stiffness** using just few seconds of data

Multi-agent Planning In Limited Communication Using Game Theory And Optimal Control
Jan 2025 – May 2025

Course: Optimal Control by Prof.Rachel Kalpana Kalaimani, IIT Madras

- Formulated the mathematical model of the multi-agent system using **differential games** and **Nash equilibrium** based approaches

- Used **Hamilton Jacobi Bellman** equations to derive the optimum control input based on a directed graph.
- Validated the model and optimal control successfully using a **simple numerical simulation in MATLAB**

Critical Comparison Of Motion Planning Algorithms For A Mobile Manipulator With A 5 DOF Arm *Jan 2025 – May 2025*

Course: Motion Planning by Prof.Nirav Patel, IIT Madras

- Implemented **A*, PRM** for the base and **RRT, RRT*** algorithm for the arm on a KUKA Youbot in a Coppeliasim simulation
- Concluded the separate planning approach for arm and base is a more efficient way as base can be planned with A* very quickly
- Used **Gradient descent** to further refine jerky paths from the RRT algorithm for the arm to ensure that it is real world deployable

Dynamic Characterization of an Underwater Flexible Single-Link Manipulator *Jul 2024 – Nov 2024*

Course: Finite Element Analysis by Prof. Sunderajan Natarajan, IIT Madras

- Developed a mathematical model of an soft robot using **Finite Element Method**, modeling it as a rotating beam with deflection
- Based on the model successfully determined the resonance frequency of the manipulator using **frequency marching simulation**

Pitch Control in a Tandem Rotor Helicopter using a PID Controller *Apr 2023 – May 2023*

ME2400 Measurements Instrumentation and Control under Dr. Manish Anand

- Developed and implemented the code for a **PID controller** in Arduino IDE to stabilize the pitch of the prototype using thrust from **2 DC motors** to any specified angle between **-30° to 30°** using **MPU6050** sensors and **Arduino UNO**

PROFESSIONAL EXPERIENCE

Teaching Assistantship - Robot Control Course *Aug 2025 – Nov 2025*

Professor :Dr. Guruprasad

- Formulated a course structure in discussion with professor from scratch as it is being offered for the first time
- Took tutorial sessions and designed assignments that will give the students a hands-on experience in the course

Product Supply Internship - High Density Liquid Detergent Production - Procter& Gamble *May 2024 – July 2024*

Guides:Mrs.Kavya Mohan & Mr.Aniket Munjal

- Reduced utility costs in detergent manufacturing by **30%** through **optimization** of heat cycles and **heat recovery systems**
- Led **material utilization improvements** in production, leading to **30 million** INR annual savings

TECHNICAL SKILLS

Languages: Python,MATLAB, C, C++, Lua LaTeX
Tools: ROS, Fusion 360, SolidWorks, Simulink, Arduino IDE, Mathematica, CoppeliaSim

RELEVANT COURSEWORK

- **Control and Planning:** Non Linear System Analysis*, Reinforcement Learning*,Data Driven Control, Optimal Control, Foundation of Machine Learning, Motion Planning, Control Systems
- **Robotics:** Introduction to Robotics, Linear Algebra, Probability, Statistics and Stochastic Process, Mechanics of Serial Manipulators, Control Systems, Field and Service Robotics,Kinematics and Dynamics of Machinery, Design of Machine Elements
- **Mathematic Preliminaries** Linear Algebra, Probability, statistics and stochastic process, Differential equations, Inverse Methods in heat transfer, Design and Optimisation of Energy Systems, Finite Element Analysis
- **Additional Certifications:**Data-Driven Learning of Dynamics for Control with Applications to Robotics and Power Systems*(* indicates Ongoing)*

SCHOLASTIC ACHIEVEMENT

- Achieved **1st rank** in **Oman** and **2nd rank** in the **Gulf** in the **Class 10 CBSE Exams** (2019) out of **14 million** students
- Attained the **highest grade** in the **Shastra Prathiba Competition** for **two consecutive years** (2017, 2018)
- Ranked in the top **0.5%** in Joint Entrance Exam-Mains 2021 (out of **1.2 million** students) and achieved an **All India Rank 3991** in Joint Entrance Exam-Advanced 2021
- Secured **66th rank** in **KEAM** (Kerala Entrance Exam) 2021 among over **80,000** students
- Awarded the **MCM (Merit Cum Means) Scholarship** for top academic performance at IIT Madras
- Stood **fourth** among around 100 teams in the National Robotics Challenge, Unacademy

EXTRACURRICULAR ACTIVITIES

Workshop Trainer, Robotics 101, Shaastra IIT Madras *Jan 2025*
Taught a comprehensive 6 hour workshop to over 120 students on Robotics Control and Simulation
Founder and Head, Sponsorship and Industrial Relations Team, CFI, IIT Madras *Sept 2023 - July 2024*
Closed sponsorships worth 20 lakhs, onboarded Stellantis as a sponsor, and orchestrated various events and their PR.
Events and Networking Manager, Entrepreneurship Cell, IIT Madras *Apr 2022 - May 2023*
Organized events for 50+ student startups, pioneered an event, and closed sponsorship, conducted a nation wide olympiad
Professionally trained in Fine Arts *2010 - Current*
Enjoys acrylic painting, has won several national level accolades in poster making for sustainability efforts